

NEXT-GEN WEB LEARNING SYSTEM

Synapse Academy

Next.js (App Router) & Tailwind CSS Line-by-Line Technical Code Explanations

Created dynamically for educational guidance and onboarding.

Target Framework: Next.js v16 & React 19 • CSS Stack: Tailwind CSS v3

1. Framework & CSS Stack Primers

Foundational concepts required before diving into the code.

Next.js App Router & React 19

- **App Router Architecture:** Folder structure dictates routes. Files inside `src/app/` map to pages. A folder named `about` containing `page.jsx` defines the `/about` endpoint.
- **Layouts & Pages:** `layout.jsx` acts as a shell that surrounds pages, helping preserve state and nesting sections.
- **Server vs Client Components:** By default, Next.js renders files as Server Components. To enable interactivity (e.g. state management, animations, events), you must specify `"use client"` at the top of the file.
- **Optimized Google Fonts:** Fonts are loaded dynamically at compile time, eliminating layout shifts (CLS).

Tailwind CSS v3 Paradigm

- **Utility-First Design:** Styling is achieved by building short, predefined utility classes (e.g. `flex`, `rounded-full`) on markup tags, keeping styles closely coupled to HTML elements.
- **Responsive Variants:** Responsive breakpoints are written as prefixes (e.g., `md:flex`). The styles apply when the viewport meets or exceeds that width.
- **State Variants:** Modifiers like `hover:`, `focus:`, or `group-hover:` allow dynamic styling without custom JS trigger scripts.
- **Arbitrary Values:** Brackets syntax (e.g. `bg-[#030014]`) lets you inject specific inline design tokens on-the-fly.

2. Detailed File Walkthroughs

Block-by-block breakdown of every configuration and source file.

globals.css (Global Stylesheet)

src/app/globals.css

This stylesheet sets up global Tailwind directives, defines theme variables, customizes scrollbars, and includes utility classes for glassmorphism and text gradients.

Lines: 1-3

```
@tailwind base;  
@tailwind components;  
@tailwind utilities;
```

EXPLANATION DETAILS

These are the three main Tailwind CSS entry directives. When compiled, Tailwind replaces these with its foundational resets (Preflight), custom component classes, and utility helper classes.

Lines: 5-29

```
@layer base {  
  :root {  
    --background: 0 0% 100%;  
    --foreground: 224 71.4% 4.1%;  
  
    --font-inter: 'Inter', sans-serif;  
    --font-outfit: 'Outfit', sans-serif;  
  }  
  
  .dark {  
    --background: 224 71.4% 4.1%;  
    --foreground: 210 20% 98%;  
  }  
  
  body {  
    background-color: #faf9f6;  
    color: #0f172a;  
    font-family: var(--font-inter);  
    overflow-x: hidden;  
  }  
}
```

```
h1, h2, h3, h4, h5, h6 {
  font-family: var(--font-outfit);
}
```

EXPLANATION DETAILS

The @layer base directive injects styles into Tailwind's base layer. It establishes CSS custom properties (variables) for backgrounds, foregrounds, and font families. It applies a default off-white background (#faf9f6), dark slate text (#0f172a), and disables horizontal scrolling (overflow-x: hidden) on the body, whilst setting headers to use the Outfit typeface.

Lines: 31-48

```
"code-comment"> /* Custom Scrollbar */
::-webkit-scrollbar {
  width: 8px;
}

::-webkit-scrollbar-track {
  background: #faf9f6;
}

::-webkit-scrollbar-thumb {
  background: #cbd5e1;
  border-radius: 4px;
}

::-webkit-scrollbar-thumb:hover {
  background: #94a3b8;
}
```

EXPLANATION DETAILS

Customizes the browser scrollbar for Webkit-based browsers (Chrome, Safari, Edge). Sets a thin scrollbar (8px) with a light background and slate/indigo theme (#cbd5e1 hovering to #94a3b8) to align with the academy theme.

Lines: 55-68

```
"code-comment">/* Glassmorphism utility */  
.glass-panel {  
  background: rgba(255, 255, 255, 0.7);  
  backdrop-filter: blur(12px);  
  -webkit-backdrop-filter: blur(12px);  
  border: 1px solid rgba(99, 102, 241, 0.08);  
}  
  
.glass-panel-heavy {  
  background: rgba(255, 255, 255, 0.9);  
  backdrop-filter: blur(20px);  
  -webkit-backdrop-filter: blur(20px);  
  border: 1px solid rgba(99, 102, 241, 0.12);  
}
```

EXPLANATION DETAILS

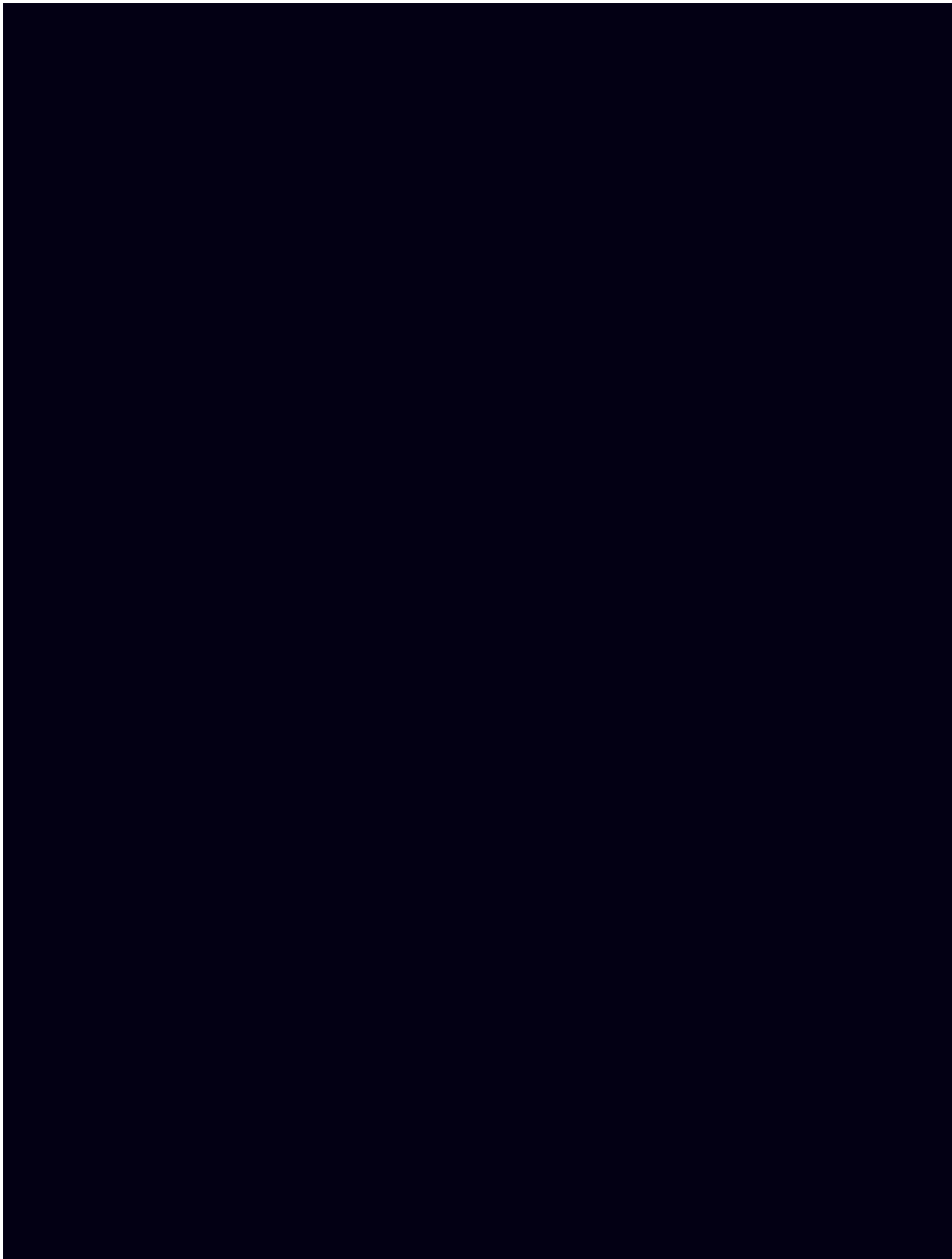
Creates a modern "frosted glass" aesthetic. It uses a very transparent white background, a CSS blur filter (12px or 20px) to blur everything underneath the panel, and a subtle semi-transparent border to simulate glass reflection.

Lines: 70-80

```
.text-gradient {  
  background: linear-gradient(to right, #0f172a, #312e81, #4f46e5);  
  -webkit-background-clip: text;  
  -webkit-text-fill-color: transparent;  
}  
  
.text-gradient-neon {  
  background: linear-gradient(to right, #4f46e5, #6366f1, #2563eb, #0891b2);  
  -webkit-background-clip: text;  
  -webkit-text-fill-color: transparent;  
}
```

EXPLANATION DETAILS

Applies visual gradient sweeps to text. It works by drawing a background linear gradient, using `background-clip: text` to clip that background to the shape of the characters, and setting `text-fill-color: transparent` to reveal the gradient underneath.



layout.jsx (Root Layout)

src/app/layout.jsx

The root layout page defines the structure (HTML and Body tags) that surrounds all pages. In Next.js App Router, this layout persists state and wraps all children nodes.

Lines: 1-15

```
import { Inter, Outfit } from "next/font/google";
import "./globals.css";

const inter = Inter({
  variable: "--font-inter",
  subsets: ["latin"],
  display: "swap",
});

const outfit = Outfit({
  variable: "--font-outfit",
  subsets: ["latin"],
  display: "swap",
});
```

EXPLANATION DETAILS

Imports Google Fonts using Next.js's optimized Font loader. The fonts are loaded with CSS variables (--font-inter, --font-outfit) and subsetting to "latin". display: "swap" instructs the browser to use a fallback font while downloading, preventing blank screens.

Lines: 16-20

```
export const metadata = {
  title: "Synapse Academy | The Creative & Tech Learning Platform",
  description: "Master creative motion design, machine learning, and advanced frontend engineering",
};
```

EXPLANATION DETAILS

Next.js metadata API. By exporting a static metadata object from a server layout, Next.js compiles these tags dynamically into the HTML head (e.g. <title>, <meta name="description">) improving SEO and sharing previews.

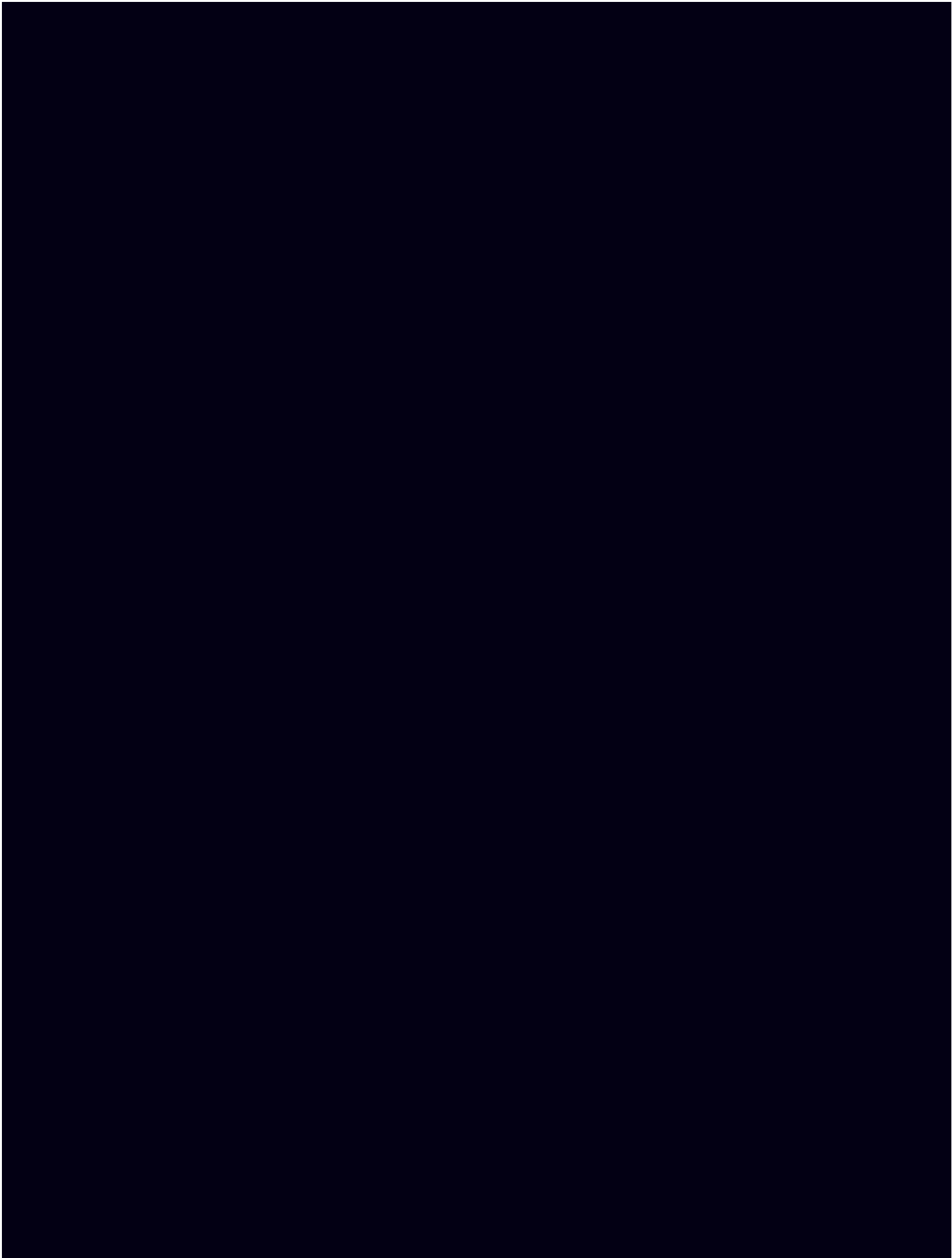
Lines: 21-32

```
"code-keyword">export "code-keyword">default "code-keyword">function RootLayout({ children }) {
  "code-keyword">return (
    <html
      lang="en"
      className={`"${inter.variable}" "${outfit.variable}" h-full antialiased scroll-smooth`}
    >
      <body className="min-h-full flex flex-col bg-[#faf9f6] text-slate-900 selection:bg-indigo-100 selection:text-indigo-900"
        {children}
      >
    </body>
  </html>
  );
}
```

EXPLANATION DETAILS

The RootLayout functional component. It accepts a children prop which is the current route page. The font variables are appended as CSS classes onto <html> so they are accessible anywhere in the CSS tree. Tailwind classes on body:

- min-h-full: min-height: 100%
- flex flex-col: vertical flexbox layout
- bg-[#faf9f6]: custom light background
- text-slate-900: dark slate text
- selection:bg-indigo-100 selection:text-indigo-900: custom highlight selection styling



page.jsx (Main Home Page)

src/app/page.jsx

The page file defines the UI for the main "/" route. It imports sections and lays them out vertically with clean background elements.

Lines: 1-8

```
"code-keyword">import Navbar "code-keyword">from "@components/Navbar";
"code-keyword">import Hero "code-keyword">from "@components/Hero";
"code-keyword">import Tracks "code-keyword">from "@components/Tracks";
"code-keyword">import Process "code-keyword">from "@components/Process";
"code-keyword">import Testimonials "code-keyword">from "@components/Testimonials";
"code-keyword">import Pricing "code-keyword">from "@components/Pricing";
"code-keyword">import CTA "code-keyword">from "@components/CTA";
"code-keyword">import Footer "code-keyword">from "@components/Footer";
```

EXPLANATION DETAILS

Imports React components using the @/ path alias configured in jsconfig.json. This alias references the src/ directory directly, avoiding messy relative paths like ../../components/...

Lines: 10-33

```
"code-keyword">export "code-keyword">default "code-keyword">function Home() {
  "code-keyword">return (
    <div className="relative flex min-h-screen flex-col overflow-hidden bg-[#faf9f6]">
      {/* Background ambient lighting */}
      <div className="absolute top-0 left-0 right-0 h-[500px] bg-gradient-to-b ">from-indi

      "code-comment">{/* Main navigation header */}
      <Navbar />

      "code-comment">{/* Main landing sections */}
      <main className="flex-grow">
        <Hero />
        <Tracks />
        <Process />
        <Testimonials />
        <Pricing />
```

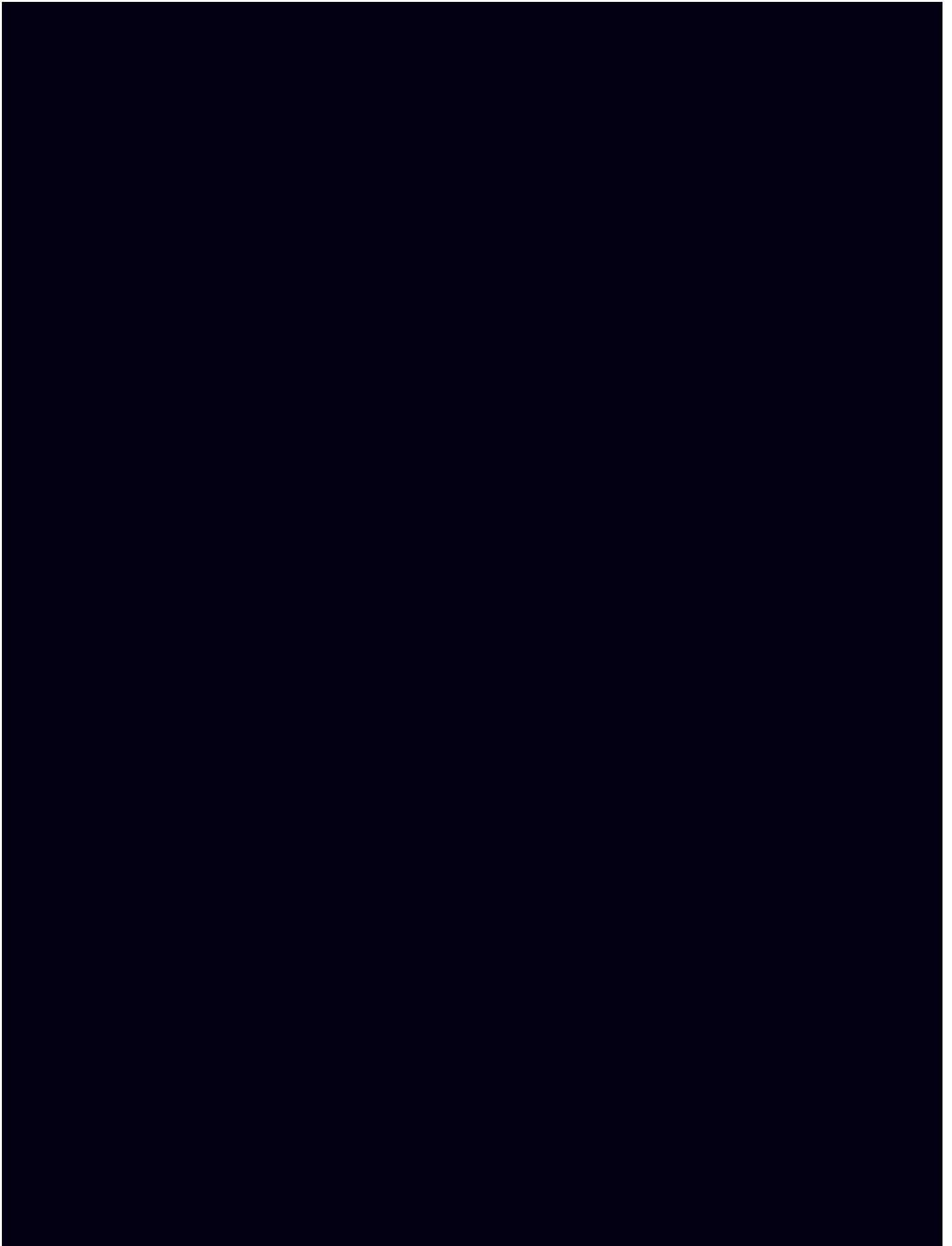
```
        <CTA />
      </main>

      "code-comment">{/* Page Footer */}
      <Footer />
    </div>
  );
}
```

EXPLANATION DETAILS

The Home component structures the page.

- relative flex min-h-screen flex-col: creates a flex container spanning at least full-height screen, positioned relative to hold absolute backgrounds.
- overflow-hidden: clamps animated components flying in from sides.
- Gradient ambient light: absolute positioned box at the top (h-[500px]) carrying a vertical fading linear gradient of Indigo (from-indigo-100/40 with 40% opacity) down to transparent. pointer-events-none allows clicks to pass through it.
- main className="flex-grow": layout expands to fill leftover space, keeping footer pushed to bottom.



Navbar.jsx (Navigation Header)

src/components/Navbar.jsx

The Navigation Header features sliding menu item overlays, Framer Motion client transitions, layout state tracking, and a collapsible mobile menu.

Lines: 1-1

```
">use client";
```

EXPLANATION DETAILS

This directive declares a "Client Component" in Next.js. By default, Next.js components are Server Components (fast, compiled on server). Adding "use client" enables browser-only capabilities like React hooks (useState, useEffect) and Framer Motion client animations.

Lines: 3-12

```
import { useState } from "react";
import { motion, AnimatePresence } from "framer-motion";
import { Sparkles, Menu, X, ArrowRight } from "lucide-react";

const navItems = [
  { name: "Tracks", href: "#tracks" },
  { name: "Curriculum", href: "#process" },
  { name: "Pricing", href: "#pricing" },
  { name: "Success Stories", href: "#testimonials" },
];
```

EXPLANATION DETAILS

Imports local state management, Framer Motion components for smooth transitions, and icons from lucide-react. Declares a static navItems array containing names and anchor hashes for fast navigation.

Lines: 14-24

```

"code-keyword">export "code-keyword">default "code-keyword">function Navbar() {
  "code-keyword">const [hoveredIndex, setHoveredIndex] = useState(null);
  "code-keyword">const [isOpen, setIsOpen] = useState("code-keyword">>false);

  "code-keyword">return (
    <motion.header
      initial={{ y: -100, opacity: 0 }}
      animate={{ y: 0, opacity: 1 }}
      transition={{ duration: 0.8, ease: "easeOut" }}
      className="fixed top-0 left-0 right-0 z-50 px-4 py-4 md:px-8"
    >

```

EXPLANATION DETAILS

Initializes states for navigation hover pill trackers and mobile slide-out toggles. Wraps the header in `<motion.header>` to execute an intro animation (sliding down from -100px on Y-axis with fading opacity). Tailwind classes:

- `fixed top-0 left-0 right-0`: locks navbar to viewport top.
- `z-50`: places it on top of all other elements.
- `px-4 py-4 md:px-8`: standard side margins scaling up to large screens.

Lines: 25-39

```

<div className="mx-auto max-w-7xl">
  <nav className="relative flex items-center justify-between rounded-full border border
    "code-comment">{/* Logo */}
    <a href="#" className="flex items-center space-x-2 text-xl font-bold font-display t
      <motion.div
        whileHover={{ rotate: 180, scale: 1.1 }}
        transition={{ type: "spring", stiffness: 200, damping: 10 }}
        className="flex h-9 w-9 items-center justify-center rounded-xl bg-gradient-to-t
      >
      <Sparkles size={18} />
    </motion.div>
    <span className="bg-gradient-to-r ">from-slate-900 to-indigo-950 bg-clip-text te
      Synapse
    </span>
  </a>

```


EXPLANATION DETAILS

Defines the inner nav shell. Tailwind classes:

- rounded-full border border-white/5 bg-slate-950/45 px-6 py-3: creates a floating capsule bar with 5% white borders and transparent backdrop.
- backdrop-blur-md: CSS backdrop-filter blur (blurs page contents passing behind the glass).
- Logo incorporates a gradient (from-indigo-500 to-cyan-500) and a custom shadow glow.
- motion.div whileHover={{ rotate: 180, scale: 1.1 }}: Framer Motion rotates the Sparkles icon 180 degrees and scales it up slightly on mouse hover.

Lines: 41-62

```
"code-comment">{/* Desktop Nav Items */}
<ul className="hidden md:flex items-center space-x-1">
  {navItems.map((item, index) => (
    <li key={item.name} className="relative">
      <a
        href={item.href}
        onMouseEnter={() => setHoveredIndex(index)}
        onMouseLeave={() => setHoveredIndex(null)}
        className="relative block px-4 py-2 text-sm font-medium text-slate-600 tran
      >
        {hoveredIndex === index && (
          <motion.span
            layoutId="navHover"
            transition={{ type: "spring", stiffness: 350, damping: 25 }}
            className="absolute inset-0 z-0 rounded-full bg-indigo-50/70 border bor
          />
        )}
        <span className="relative z-10">{item.name}</span>
      </a>
    </li>
  )]}
</ul>
```

EXPLANATION DETAILS

Desktop link list. Classes "hidden md:flex" hide the list on mobile and show it as a horizontal flexbox on medium screens and up. Uses a brilliant UX micro-interaction: a shared layout animations tab (layoutId="navHover"). When switching hovered links, Framer Motion automatically animates the background pill container sliding from the old item to the new item, instead of fading out and in.

Lines: 64-84

```

"code-comment">>{/* CTA Button */}
<div className="hidden md:block">
  <motion.a
    href="#pricing"
    whileHover={{ scale: 1.05, boxShadow: "0px 6px 20px rgba(99, 102, 241, 0.25)" }}
    whileTap={{ scale: 0.95 }}
    className="inline-flex items-center space-x-2 rounded-full bg-gradient-to-r "
  >
    <span>Enroll Now</span>
    <ArrowRight size={14} />
  </motion.a>
</div>

"code-comment">>{/* Mobile Menu Toggle */}
<button
  onClick={() => setIsOpen(!isOpen)}
  className="block md:hidden rounded-lg p-1 text-slate-600 hover:text-indigo-600 focus:outline-none"
>
  {isOpen ? <X size={24} /> : <Menu size={24} />}
</button>
</nav>

```

EXPLANATION DETAILS

CTA Button uses Framer Motion whileHover to increase size and emit a shadow glow on hover, and whileTap to scale down. Mobile menu toggle is built using "block md:hidden" (displays on mobile, hides on desktop). Clicking the button flips the isOpen boolean state, changing the icon between Menu and Close.

Lines: 87-96

```

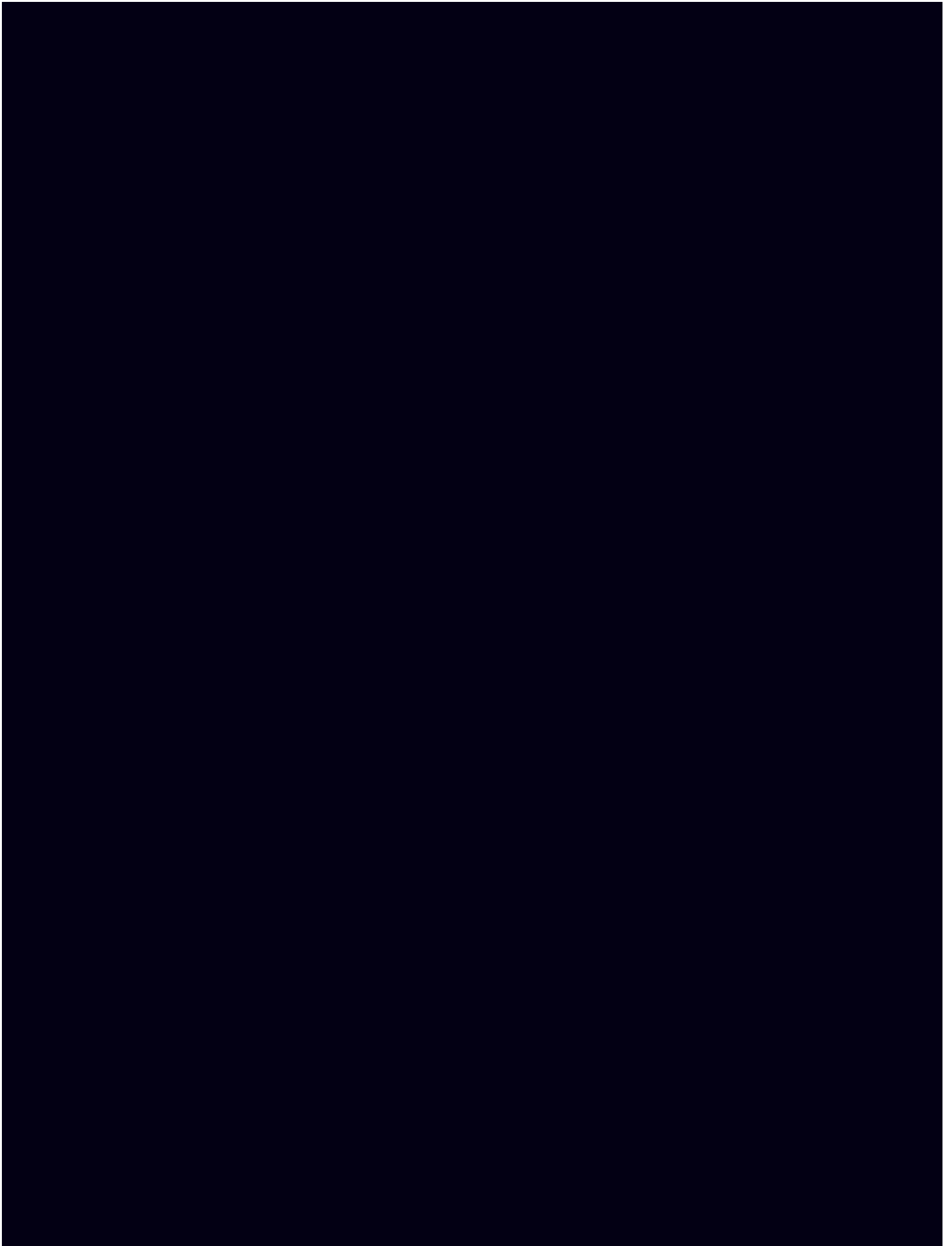
"code-comment">>{/* Mobile Drawer */}
<AnimatePresence>
  {isOpen && (
    <motion.div
      initial={{ opacity: 0, height: 0 }}
      animate={{ opacity: 1, height: "auto" }}
      exit={{ opacity: 0, height: 0 }}
      transition={{ duration: 0.3 }}
    >

```

```
className="absolute left-4 right-4 top-24 z-40 overflow-hidden rounded-2xl border"
```

EXPLANATION DETAILS

Handles the mobile drawer menu. `AnimatePresence` is a specialized Framer Motion component that enables elements to animate *before* they are completely removed from the React DOM tree. When `isOpen` is toggled off, the drawer slides up before unmounting.



Hero.jsx (Hero Section)

src/components/Hero.jsx

The landing cover section of the website. It contains animated gradient background spheres, floating content cards, microstats, and an overlay video modal popup.

Lines: 1-9

```
"">use client";

"code-keyword">import { useState } "code-keyword">from "react";
"code-keyword">import { motion, AnimatePresence } "code-keyword">from "framer-motion";
"code-keyword">import { Play, Sparkles, BookOpen, Star, Activity, X } "code-keyword">from "lucide-react";

"code-keyword">export "code-keyword">default "code-keyword">function Hero() {
  "code-keyword">const [isVideoOpen, setIsVideoOpen] = useState("code-keyword">>false);
```

EXPLANATION DETAILS

Declares client rendering and imports Framer Motion, standard React state hook, and icons from lucide-react.

Lines: 10-28

```
"code-keyword">const containerVariants = {
  hidden: { opacity: 0 },
  visible: {
    opacity: 1,
    transition: {
      staggerChildren: 0.2,
      delayChildren: 0.1,
    },
  },
};

"code-keyword">const itemVariants = {
  hidden: { y: 30, opacity: 0 },
  visible: {
    y: 0,
```

```
    opacity: 1,  
    transition: { type: "spring", stiffness: 100, damping: 15 },  
  },  
};
```

EXPLANATION DETAILS

Defines animation variants for orchestration. By applying `staggerChildren: 0.2` on the container, children elements carrying the `itemVariants` configuration will fade and slide up sequentially (0.2s spacing) rather than animating all at once, creating a premium presentation flow.

Lines: 30-59

```
"code-keyword">const floatCards = [  
  {  
    title: "Interactive Motion",  
    subtitle: "Framer Motion & WebGL",  
    icon: <Activity className="text-cyan-600" size={18} />,  
    color: ">from-white to-cyan-50/30",  
    border: "border-cyan-200",  
    delay: 0,  
    position: "top-4 left-6 md:top-12 md:left-24",  
  },  
  {  
    title: "Machine Learning",  
    subtitle: "Neural Nets & Agents",  
    icon: <Sparkles className="text-purple-600" size={18} />,  
    color: ">from-white to-purple-50/30",  
    border: "border-purple-200",  
    delay: 1.5,  
    position: "bottom-12 right-6 md:bottom-20 md:right-16",  
  },  
  {  
    title: "Frontend Engineering",  
    subtitle: "Next.js & React 19",  
    icon: <BookOpen className="text-indigo-600" size={18} />,  
    color: ">from-white to-indigo-50/30",  
    border: "border-indigo-200",  
    delay: 0.7,  
    position: "top-1/3 right-12 md:top-1/3 md:right-32",  
  },  
];
```

EXPLANATION DETAILS

Initializes a dataset of floating information cards including metadata like icons, colors, relative offsets, and delay factors to make cards drift independently.

Lines: 60-93

```
"code-keyword">return (  
  <section className="relative min-h-screen w-full overflow-hidden bg-[#faf9f6] pt-28 pb-16  
    "code-comment">{/* Background Gradients */}  
    <div className="absolute inset-0 z-0">  
      <motion.div  
        animate={{  
          scale: [1, 1.2, 1],  
          x: [0, 50, 0],  
          y: [0, -30, 0],  
        }}  
        transition={{  
          duration: 15,  
          repeat: Infinity,  
          ease: "easeInOut",  
        }}  
        className="absolute top-1/4 left-1/4 h-[350px] w-[350px] rounded-full bg-indigo-100  
      />  
      <motion.div  
        animate={{  
          scale: [1, 1.1, 1],  
          x: [0, -40, 0],  
          y: [0, 40, 0],  
        }}  
        transition={{  
          duration: 12,  
          repeat: Infinity,  
          ease: "easeInOut",  
        }}  
        className="absolute bottom-1/4 right-1/4 h-[400px] w-[400px] rounded-full bg-blue-1  
      />  
    "code-comment">{/* Dot Grid */}  
    <div className="absolute inset-0 bg-[linear-gradient(to_right,#00000004_1px,transparent  
  </div>
```

EXPLANATION DETAILS

Sets up the interactive background layer:

- Two background spheres are animated to drift and scale slowly in infinite loops (repeat: Infinity) to mimic organic breathing canvas elements.
- blur-[120px]: high blur-radius values transform simple solid colored circles into ambient lights.
- The dot grid is a CSS linear background pattern clipped with a CSS mask image (radial-gradient), fading the grid at the corners so it only renders in the center page.

Lines: 98-130

```
<motion.div
  variants={containerVariants}
  initial="hidden"
  animate="visible"
  className="lg:col-span-7 text-center lg:text-left space-y-6"
>
  <motion.div
    variants={itemVariants}
    className="inline-flex items-center space-x-2 rounded-full border border-indigo-600"
  >
    <Star size={12} className="fill-indigo-600 text-indigo-600" />
    <span>Next-Gen Creative Technical Academy</span>
  </motion.div>

  <motion.h1
    variants={itemVariants}
    className="text-4xl sm:text-5xl md:text-6xl font-extrabold font-display leading-tight"
  >
    Learn in{" "}
    <span className="bg-gradient-to-r from-indigo-600 via-purple-600 to-blue-600" style="background-size: 400% 100%; background-position: 0% 0%; background-repeat: no-repeat; border-radius: 10px; padding: 2px 10px; display: inline-block; text-align: center; font-weight: bold;">Flow State</span>
    </span>
    . Master the Code.
  </motion.h1>

  <motion.p
    variants={itemVariants}
    className="text-base sm:text-lg text-slate-600 max-w-xl mx-auto lg:mx-0 leading-tight"
  >
    Unlock immersive, curriculum-led learning tracks featuring interactive code sar
  </motion.p>

  "code-comment">{/* Buttons */}
```


EXPLANATION DETAILS

Layout columns container. Using "grid grid-cols-1 lg:grid-cols-12" builds a responsive layout system (stacked on mobile, side-by-side 12-column layout on large screens). The left column spans 7 grid spaces.

Lines: 131-156

```

<motion.div
  variants={itemVariants}
  className="flex flex-col sm:flex-row items-center justify-center lg:justify-sta
>
  <motion.a
    href="#pricing"
    whileHover={{ scale: 1.05 }}
    whileTap={{ scale: 0.95 }}
    className="w-full sm:w-auto text-center rounded-full bg-gradient-to-r ">from-
  >
    Get Started Free
  </motion.a>

  <motion.button
    onClick={() => setIsVideoOpen("code-keyword">true)}
    whileHover={{ scale: 1.05, backgroundColor: "rgba(0,0,0,0.02)" }}
    whileTap={{ scale: 0.95 }}
    className="w-full sm:w-auto inline-flex items-center justify-center space-x-2
  >
    <div className="flex h-6 w-6 items-center justify-center rounded-full bg-indi
      <Play size={10} className="ml-0.5 fill-white" />
    </div>
    <span>Watch Platform Intro</span>
  </motion.button>
</motion.div>

```

EXPLANATION DETAILS

Contains CTA controls. "Get Started Free" button features a vibrant gradient with high shadow properties, while "Watch Platform Intro" loads a client modal overlay.

Lines: 157-178

```

"code-comment">{/* Microstats */}
<motion.div
  variants={itemVariants}
  className="flex items-center justify-center lg:justify-start space-x-8 pt-8 bor
>
  <div>
    <p className="text-2xl font-bold text-slate-900 font-display">12,000+</p>
    <p className="text-xs text-slate-500">Active Students</p>
  </div>
  <div className="h-8 w-px bg-slate-200" />
  <div>
    <p className="text-2xl font-bold text-slate-900 font-display">98.7%</p>
    <p className="text-xs text-slate-500">Completion Rate</p>
  </div>
  <div className="h-8 w-px bg-slate-200" />
  <div>
    <p className="text-2xl font-bold text-slate-900 font-display">400+</p>
    <p className="text-xs text-slate-500">Video Projects</p>
  </div>
</motion.div>
</motion.div>

```

EXPLANATION DETAILS

Renders microstats metrics separated by vertical border lines. Margins and padding scale responsively, shifting alignment based on screen widths.

Lines: 180-230

```

<div className="lg:col-span-5 relative min-h-[400px] md:min-h-[480px] w-full fle
"code-comment">{/* Center Glowing Hub */}
<motion.div
  animate={{
    scale: [0.9, 1.1, 0.9],
    opacity: [0.5, 0.8, 0.5],
  }}
  transition={{
    duration: 6,
    repeat: Infinity,
    ease: "easeInOut",
  }}
  className="absolute h-56 w-56 rounded-full bg-indigo-100/40 blur-[80px]"
/>

```

```

"code-comment">>{/* Orbit paths */}
<div className="absolute h-80 w-80 rounded-full border border-dashed border-slate-400" style={{
  top: 0, left: 0, right: 0, bottom: 0;
}}>
<div className="absolute h-[420px] w-[420px] rounded-full border border-dashed border-slate-400" style={{
  top: 0, left: 0, right: 0, bottom: 0;
}}>

"code-comment">>{/* Animated Floating Cards */}
{floatCards.map((card, idx) => (
  <motion.div
    key={idx}
    animate={{
      y: [0, -12, 0],
    }}
    transition={{
      duration: 6 + idx * 1.5,
      repeat: Infinity,
      ease: "easeInOut",
      delay: card.delay,
    }}
    whileHover={{
      scale: 1.05,
      y: -15,
      boxShadow: "0px 10px 30px rgba(99, 102, 241, 0.08)",
      borderColor: "rgba(99, 102, 241, 0.3)",
    }}
    className={`absolute ${card.position} z-20 flex items-center space-x-3.5 rounded-xl`}
  >
    <div className="flex h-10 w-10 items-center justify-center rounded-xl bg-slate-100" style={{
      width: 40px, height: 40px;
    }}>
      {card.icon}
    </div>
    <div>
      <h3 className="text-sm font-bold text-slate-900 font-display">{card.title}</h3>
      <p className="text-xs text-slate-500">{card.subtitle}</p>
    </div>
  </motion.div>
)
)}}
</div>

```

EXPLANATION DETAILS

Structures the floating card grid dashboard. Uses `lg:col-span-5` to assign 5 units out of the 12-column grid to this section. Loops through the `floatCards` array. Each card is absolutely positioned and runs a drifting animation (`y: [0, -12, 0]`) with dynamic speeds (`6 + idx * 1.5` seconds) so the cards float independently.

Lines: 234-295

```

"code-comment">{/* Video Modal Overlay */}
<AnimatePresence>
  {isVideoOpen && (
    <motion.div
      initial={{ opacity: 0 }}
      animate={{ opacity: 1 }}
      exit={{ opacity: 0 }}
      className="fixed inset-0 z-50 flex items-center justify-center bg-slate-900/60
    >
    <motion.div
      initial={{ scale: 0.9, opacity: 0 }}
      animate={{ scale: 1, opacity: 1 }}
      exit={{ scale: 0.9, opacity: 0 }}
      transition={{ type: "spring", damping: 20 }}
      className="relative w-full max-w-4xl overflow-hidden rounded-2xl border border
    >
    "code-comment">{/* Close Button */}
    <button
      onClick={() => setIsVideoOpen("code-keyword">false)}
      className="absolute top-4 right-4 z-10 flex h-8 w-8 items-center justify-c
    >
      <X size={18} />
    </button>

    "code-comment">{/* Simulated Video Player aspect-video */}
    <div className="relative aspect-video w-full bg-slate-50 border border-slate
      "code-comment">{/* Custom glowing particle nodes in preview */}
      <div className="absolute top-10 left-10 h-2 w-2 rounded-full bg-cyan-400 b
      <div className="absolute bottom-10 right-10 h-3 w-3 rounded-full bg-purple

    <div className="relative flex flex-col items-center space-y-4 text-center
      <div className="flex h-16 w-16 items-center justify-center rounded-full
        <Sparkles size={28} className="animate-spin-slow" />
      </div>
      <h3 className="text-2xl font-bold font-display text-slate-900">Synapse A
      <p className="text-sm text-slate-500 max-w-md">
        Take interactive code tests, compile motion graphics live in the brows
      </p>
      <button
        onClick={() => setIsVideoOpen("code-keyword">false)}
        className="rounded-full bg-indigo-600 px-6 py-2 text-xs font-semibold
      >
        Close Demo
      </button>
    </div>

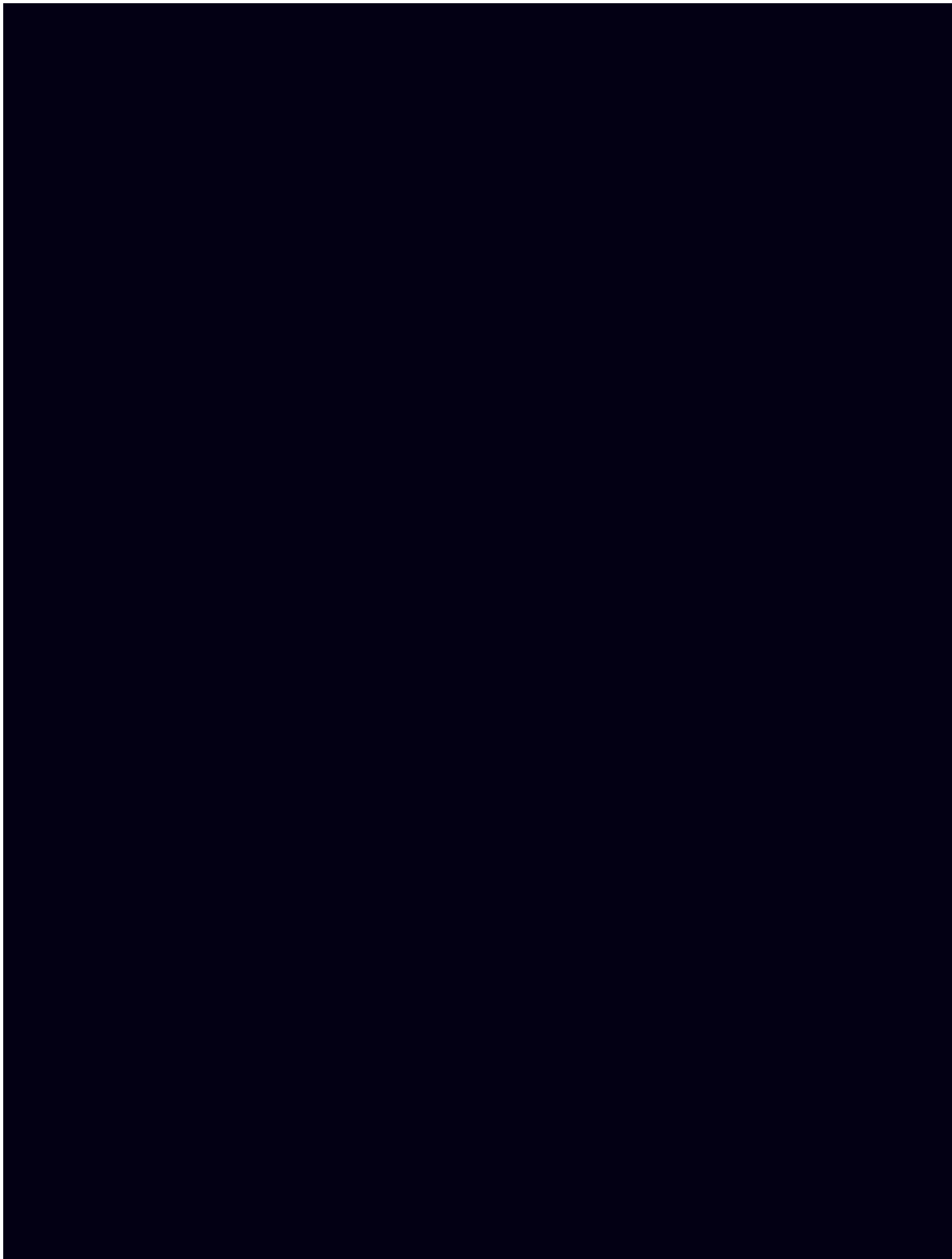
    "code-comment">{/* Simulated playback bar */}
    <div className="absolute bottom-0 left-0 right-0 h-1.5 bg-slate-200">

```

```
        <motion.div
          initial={{ width: 0 }}
          animate={{ width: "65%" }}
          transition={{ duration: 15, repeat: Infinity, ease: "linear" }}
          className="h-full bg-gradient-to-r from-indigo-600 to-violet-600"
        />
      </div>
    </div>
  </motion.div>
</motion.div>
  )}
</AnimatePresence>
</section>
);
```

EXPLANATION DETAILS

Implements the full screen overlay demo video popup. Uses AnimatePresence to animate the backdrop and card transitions smoothly when the modal is closed.



Tracks.jsx (Learning Pathways)

src/components/Tracks.jsx

The curriculum explorer page contains interactive category filtering tabs and responsive course catalog cards that dynamically react to mouse hover states.

Lines: 1-6

```
"">use client";

"code-keyword">import { useState } "code-keyword">from "react";
"code-keyword">import { motion, AnimatePresence } "code-keyword">from "framer-motion";
"code-keyword">import { Play, Sparkles, BookOpen, Clock, Users, ArrowRight, Zap } "code-keyw
```

EXPLANATION DETAILS

Sets up client hooks, motion animations, and icons from lucide-react.

Lines: 7-144

```
"code-keyword">const tracksData = {
  motion: [
    {
      id: "m-1",
      title: "Interactive Web Animation with Framer Motion",
      duration: "14 hrs",
      lessons: 28,
      students: "3.4k",
      difficulty: "Intermediate",
      gradient: "">from-purple-500/10 via-indigo-500/5 to-transparent",
      color: "text-purple-600 border-purple-100 bg-purple-50/50",
    },
    {
      id: "m-2",
      title: "Introduction to Three.js & 3D WebGL Spaces",
      duration: "20 hrs",
      lessons: 35,
      students: "2.1k",
      difficulty: "Advanced",
```

```
    gradient: ">from-cyan-500/10 via-blue-500/5 to-transparent",
    color: "text-cyan-600 border-cyan-100 bg-cyan-50/50",
  },
  {
    id: "m-3",
    title: "SVG Animation Mechanics & Micro-interactions",
    duration: "8 hrs",
    lessons: 14,
    students: "1.8k",
    difficulty: "Beginner",
    gradient: ">from-pink-500/10 via-rose-500/5 to-transparent",
    color: "text-pink-600 border-pink-100 bg-pink-50/50",
  },
],
frontend: [
  {
    id: "f-1",
    title: "Advanced React 19 Ecosystem & Server Actions",
    duration: "18 hrs",
    lessons: 30,
    students: "4.2k",
    difficulty: "Advanced",
    gradient: ">from-blue-500/10 via-cyan-500/5 to-transparent",
    color: "text-blue-600 border-blue-100 bg-blue-50/50",
  },
  {
    id: "f-2",
    title: "Next.js Full-Stack Architecture & App Router",
    duration: "22 hrs",
    lessons: 45,
    students: "5.1k",
    difficulty: "Intermediate",
    gradient: ">from-indigo-500/10 via-purple-500/5 to-transparent",
    color: "text-indigo-600 border-indigo-100 bg-indigo-50/50",
  },
  {
    id: "f-3",
    title: "CSS Architecture: Custom Properties & Tailwind Core",
    duration: "10 hrs",
    lessons: 18,
    students: "2.9k",
    difficulty: "Beginner",
    gradient: ">from-teal-500/10 via-emerald-500/5 to-transparent",
    color: "text-teal-600 border-teal-100 bg-teal-50/50",
  },
],
ai: [
  {
    id: "a-1",
    title: "Introduction to LLMs, Neural Nets, & Transformers",
```



```
    duration: "25 hrs",
    lessons: 40,
    students: "3.9k",
    difficulty: "Intermediate",
    gradient: ">from-amber-500/10 via-orange-500/5 to-transparent",
    color: "text-amber-700 border-amber-100 bg-amber-50/50",
  },
  {
    id: "a-2",
    title: "Building Autonomous Agents with LangChain",
    duration: "16 hrs",
    lessons: 24,
    students: "1.5k",
    difficulty: "Advanced",
    gradient: ">from-red-500/10 via-orange-500/5 to-transparent",
    color: "text-red-600 border-red-100 bg-red-50/50",
  },
  {
    id: "a-3",
    title: "Prompt Engineering & Vector Databases",
    duration: "9 hrs",
    lessons: 15,
    students: "2.6k",
    difficulty: "Beginner",
    gradient: ">from-yellow-500/10 via-amber-500/5 to-transparent",
    color: "text-amber-600 border-yellow-200 bg-yellow-50/50",
  },
],
ux: [
  {
    id: "u-1",
    title: "Interactive Prototype Design in Figma",
    duration: "12 hrs",
    lessons: 20,
    students: "2.7k",
    difficulty: "Beginner",
    gradient: ">from-fuchsia-500/10 via-purple-500/5 to-transparent",
    color: "text-fuchsia-600 border-fuchsia-100 bg-fuchsia-50/50",
  },
  {
    id: "u-2",
    title: "Design Systems & High-Fidelity Styling Handshake",
    duration: "15 hrs",
    lessons: 25,
    students: "1.9k",
    difficulty: "Intermediate",
    gradient: ">from-emerald-500/10 via-teal-500/5 to-transparent",
    color: "text-emerald-600 border-emerald-100 bg-emerald-50/50",
  },
  {
```

```

      id: "u-3",
      title: "Cognitive UX: Psychology behind User Retention",
      duration: "14 hrs",
      lessons: 22,
      students: "2.2k",
      difficulty: "Advanced",
      gradient: ">from-cyan-500/10 via-indigo-500/5 to-transparent",
      color: "text-cyan-600 border-cyan-100 bg-cyan-50/50",
    },
  ],
};

```

```

"code-keyword">const tabList = [
  { id: "motion", name: "Creative Motion" },
  { id: "frontend", name: "Frontend Architect" },
  { id: "ai", name: "AI Agents" },
  { id: "ux", name: "UX/UI Craft" },
];

```

EXPLANATION DETAILS

Declares catalog path data structured by key pathways (Motion, Frontend, AI, UX). Each course object carries dynamic properties like gradients, labels, and text colors which will be rendered in the grid.

Lines: 145-170

```

"code-keyword">export "code-keyword">default "code-keyword">function Tracks() {
  "code-keyword">const [activeTab, setActiveTab] = useState("motion");
  "code-keyword">const [hoveredCourseId, setHoveredCourseId] = useState(null);

  "code-keyword">return (
    <section id="tracks" className="relative py-24 bg-slate-50">
      { /* Glow Backdrops */ }
      <div className="absolute top-1/3 left-0 h-96 w-96 rounded-full bg-cyan-100/20 blur-[100px]" />
      <div className="absolute bottom-1/3 right-0 h-96 w-96 rounded-full bg-purple-100/25 blur-[100px]" />

      <div className="relative z-10 mx-auto max-w-7xl px-4 md:px-8">

        "code-comment">{ /* Section Header */ }
        <div className="text-center max-w-3xl mx-auto space-y-4 mb-16">
          <div className="inline-flex items-center space-x-1.5 rounded-full border border-cyan-100 px-4 py-2">
            <Zap size={12} />
            <span>Interactive Curriculum Explorer</span>
          </div>
          <h2 className="text-3xl sm:text-4xl font-extrabold font-display tracking-tight">

```

```

    Choose Your Learning Specialization
  </h2>
  <p className="text-sm sm:text-base text-slate-600">
    Switch between pathways to explore hands-on tracks. Each specialization includes
  </p>
</div>

```

EXPLANATION DETAILS

Creates the core tracks component skeleton with local active tab state tracking.

Lines: 172-194

```

<div className="flex items-center justify-center mb-12">
  <div className="flex flex-wrap items-center justify-center gap-2 rounded-full border"
    {tabList.map((tab) => (
      <button
        key={tab.id}
        onClick={() => setActiveTab(tab.id)}
        className={`relative px-5 py-2.5 rounded-full text-sm font-semibold transiti
          activeTab === tab.id ? "text-white" : "text-slate-600 hover:text-slate-900"
        }`}
      >
        {activeTab === tab.id && (
          <motion.div
            layoutId="activeTrackTab"
            className="absolute inset-0 bg-gradient-to-r ">from-indigo-600 to-violet-
            transition={{ type: "spring", stiffness: 380, damping: 30 }}
          />
        )}
        <span className="relative z-10">{tab.name}</span>
      </button>
    ))}
  </div>
</div>

```

EXPLANATION DETAILS

The tab switcher center navigation. Uses activeTab local state. Highlights the selected tab using Framer Motion <motion.div layoutId="activeTrackTab">. This matches layout IDs across renders,

prompting the background purple pill gradient to slide smoothly to the newly selected button upon state changes.

Lines: 196-276

```
<div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-8">
  <AnimatePresence mode="popLayout">
    {tracksData[activeTab].map((course, index) => (
      <motion.div
        key={course.id}
        layout
        initial={{ opacity: 0, y: 20 }}
        animate={{ opacity: 1, y: 0 }}
        exit={{ opacity: 0, y: -20 }}
        transition={{ duration: 0.4, delay: index * 0.05 }}
        onMouseEnter={() => setHoveredCourseId(course.id)}
        onMouseLeave={() => setHoveredCourseId(null)}
        className="group relative flex flex-col justify-between h-full rounded-2xl
      >
        "code-comment">{/* Dynamic Gradient Backdrop Glow */}
        <div className={`absolute inset-0 z-0 bg-gradient-to-br ${course.gradient}`}
        <div className="absolute top-0 left-0 right-0 h-[1px] bg-gradient-to-r ">f

        "code-comment">{/* Card Content */}
        <div className="relative z-10 space-y-6">
          "code-comment">{/* Badge Row */}
          <div className="flex items-center justify-between">
            <span className={`text-xs font-bold font-display uppercase tracking-wid
              {course.difficulty}
            </span>
            <div className="flex items-center space-x-1 text-slate-500 text-xs font
              <Users size={12} />
              <span>{course.students}</span>
            </div>
          </div>

          "code-comment">{/* Title */}
          <h3 className="text-lg font-bold font-display text-slate-900 group-hover
            {course.title}
          </h3>
        </div>

        "code-comment">{/* Footer Metrics & Actions */}
        <div className="relative z-10 pt-6 mt-6 border-t border-slate-100 space-y-4
          <div className="flex items-center justify-between text-xs text-slate-600
            <div className="flex items-center space-x-1">
```

```

        <Clock size={12} className="text-slate-400" />
        <span>{course.duration}</span>
      </div>
      <div className="flex items-center space-x-1">
        <BookOpen size={12} className="text-slate-400" />
        <span>{course.lessons} Modules</span>
      </div>
    </div>

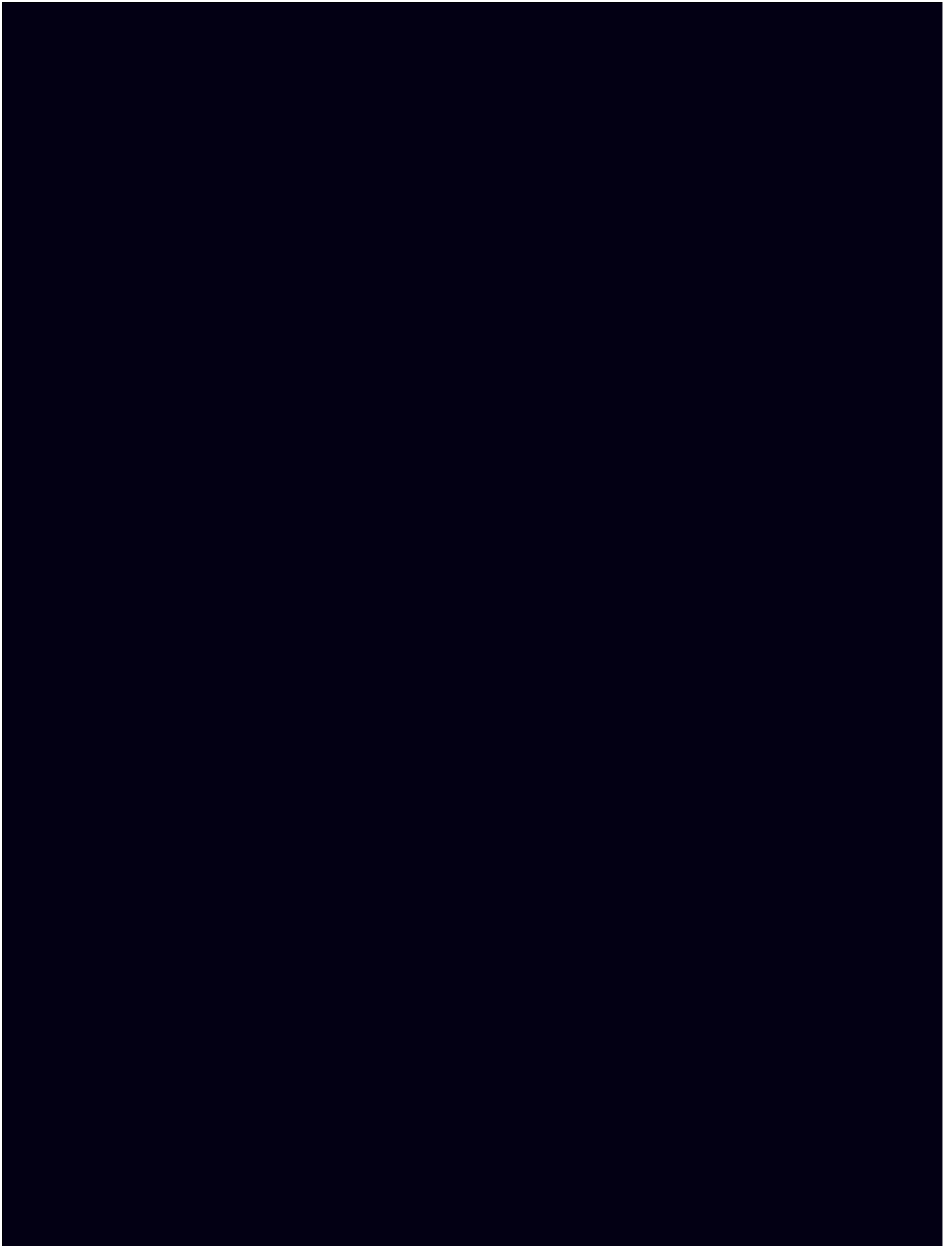
    "code-comment">{/* Dynamic Progress micro-interaction */}
    <div className="space-y-1">
      <div className="flex items-center justify-between text-[10px] text-slate-400">
        <span>Curriculum Depth</span>
        <span>100% Core</span>
      </div>
      <div className="h-1 w-full bg-slate-100 rounded-full overflow-hidden">
        <motion.div
          initial={{ width: 0 }}
          animate={{ width: hoveredCourseId === course.id ? "100%" : "30%" }}
          transition={{ duration: 0.8, ease: "easeOut" }}
          className="h-full bg-gradient-to-r from-indigo-600 to-cyan-600"
        />
      </div>
    </div>

    "code-comment">{/* Enroll button transition */}
    <motion.div
      animate={{ x: hoveredCourseId === course.id ? 4 : 0 }}
      className={`inline-flex items-center space-x-1.5 text-sm font-semibold
        hoveredCourseId === course.id ? "text-cyan-600" : "text-indigo-600"
      }`}
    >
      <span>View Curriculum</span>
      <ArrowRight size={14} />
    </motion.div>
  </div>
</motion.div>
)}}
</AnimatePresence>
</div>

```

EXPLANATION DETAILS

The courses display grid. Tailwind grid `grid-cols-1 md:grid-cols-2 lg:grid-cols-3` handles full responsive wrapping. Wrapped in `AnimatePresence mode="popLayout"`. When you toggle tabs, the courses animate out of the layout (`exit={{ opacity: 0, y: -20 }}`) and the remaining items rearrange themselves smoothly via the `layout` prop.



Process.jsx (Steps & Methodology)

src/components/Process.jsx

Visualizes the academy learning roadmap with connective linear timeline flow structures.

Lines: 1-43

```
"">use client";

"code-keyword">import { motion } "code-keyword">from "framer-motion";
"code-keyword">import { Compass, Code2, Users2, Award, ArrowUpRight } "code-keyword">from

"code-keyword">const steps = [
  {
    number: "01",
    title: "Choose Your Path",
    desc: "Pick ">from custom tracks designed by industry professionals in WebGL, React 19
    icon: <Compass className="text-indigo-600" size={24} />,
    color: "">from-indigo-100/50 to-purple-50/50",
    glow: "shadow-[0_4px_12px_rgba(99,102,241,0.06)]",
    details: ["Tailored diagnostic test", "Dedicated path counselors", "Lifetime syllabus"],
  },
  {
    number: "02",
    title: "Build in Sandbox Environments",
    desc: "Write code and render complex animations side-by-side using our built-in browser
    icon: <Code2 className="text-cyan-600" size={24} />,
    color: "">from-cyan-100/50 to-blue-50/50",
    glow: "shadow-[0_4px_12px_rgba(6,182,212,0.06)]",
    details: ["Live canvas renders", "Autosaved code checkpoints", "Hot module reloading in
  },
  {
    number: "03",
    title: "Real-time AI Co-pilot Review",
    desc: "Receive immediate syntax corrections, design improvements, and performance tips
    icon: <Users2 className="text-emerald-600" size={24} />,
    color: "">from-emerald-100/50 to-teal-50/50",
    glow: "shadow-[0_4px_12px_rgba(16,185,129,0.06)]",
    details: ["24/7 code explanation", "Performance complexity checker", "Refactoring sugg
  },
  {
    number: "04",
    title: "Earn Dynamic On-Chain Credentials",
    desc: "Publish your live project portfolios, receive verified peer-reviews, and earn c
    icon: <Award className="text-fuchsia-600" size={24} />,
  },
];
```



```

color: ">from-fuchsia-100/50 to-pink-50/50",
glow: "shadow-[0_4px_12px_rgba(217,70,239,0.06)]",
details: ["Custom portfolio website link", "Cryptographic signature", "Partner hiring
},
];

```

EXPLANATION DETAILS

Sets up local data array defining the four major roadmap steps. Establishes color themes, glow variables, and sub-details lists.

Lines: 67-83

```

"code-comment">{/* Timeline Flow */}
<div className="relative grid grid-cols-1 md:grid-cols-2 lg:grid-cols-4 gap-8">
  {steps.map((step, idx) => (
    <motion.div
      key={step.number}
      initial={{ opacity: 0, y: 40 }}
      whileInView={{ opacity: 1, y: 0 }}
      viewport={{ once: "code-keyword">true, margin: "-100px" }}
      transition={{ duration: 0.7, delay: idx * 0.15, type: "spring", stiffness: 80 }}
      whileHover={{ y: -8 }}
      className="relative flex flex-col justify-between rounded-2xl border border-slate-200
    >
    "code-comment">{/* Connector line (desktop only) */}
    {idx < steps.length - 1 && (
      <div className="hidden lg:block absolute top-12 left-[calc(100%-12px)] w-[calc(100%-12px)]
    )}
  )}

```

EXPLANATION DETAILS

Builds step card items inside a 4-column layout (lg:grid-cols-4).
 - Uses absolute positioning to inject a dashed horizontal timeline link line (hidden lg:block absolute ...) between adjacent cards, hiding the divider for the last step card.

Lines: 85-115

```

<div className="relative z-10 flex items-center justify-between mb-8">
  <div className="text-3xl font-extrabold font-display text-slate-900/10 group-
    {step.number}
  </div>
  <div className={`flex h-12 w-12 items-center justify-center rounded-2xl bg-gr
    {step.icon}
  </div>
</div>

"code-comment">{/* Description */}
<div className="relative z-10 space-y-3 mb-6">
  <h3 className="text-lg font-bold font-display text-slate-900 group-hover:text
    {step.title}
  </h3>
  <p className="text-sm text-slate-600 leading-relaxed">
    {step.desc}
  </p>
</div>

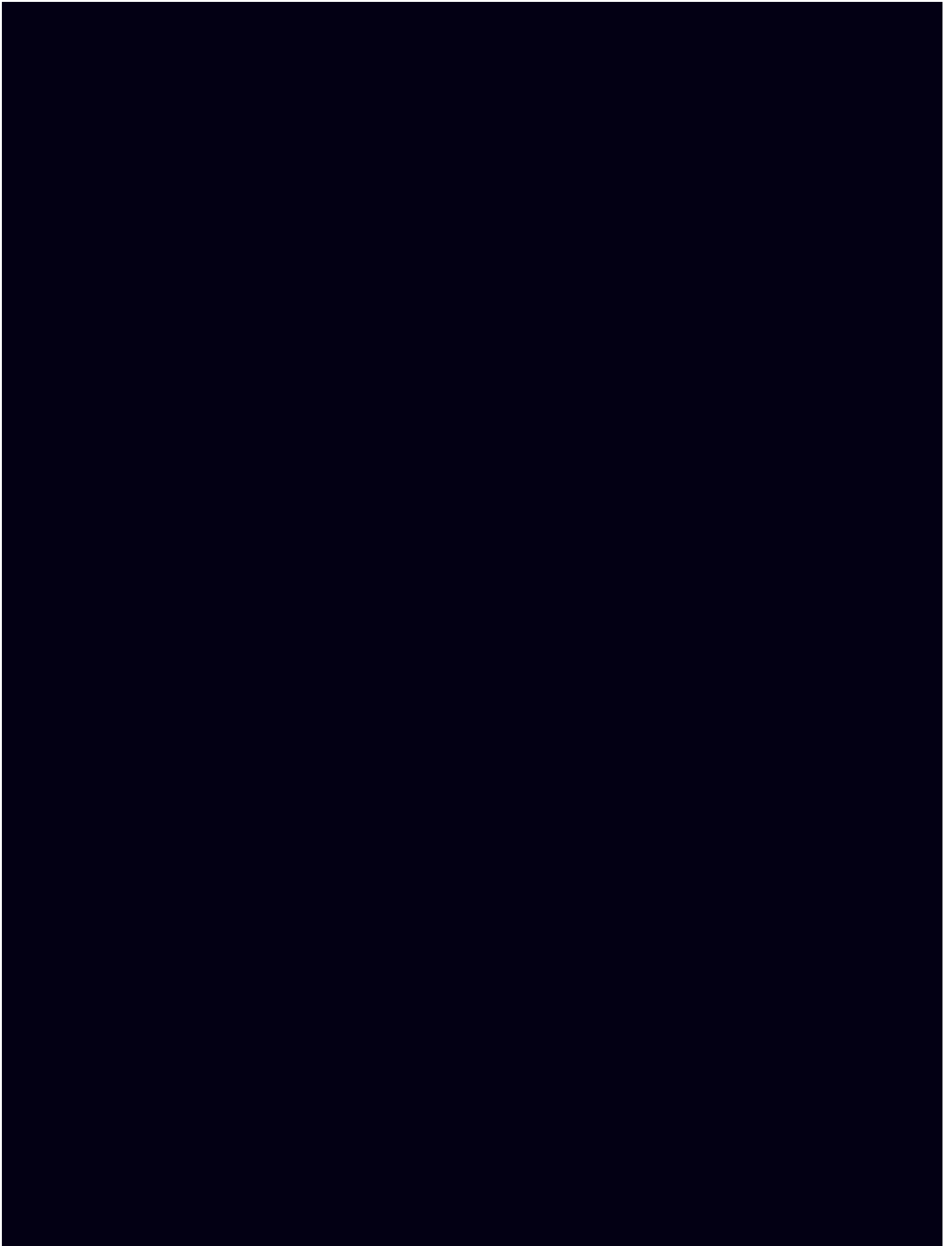
"code-comment">{/* Hover Bullet Items drawer */}
<div className="relative z-10 pt-4 border-t border-slate-100 space-y-2.5">
  {step.details.map((detail, dIdx) => (
    <div key={dIdx} className="flex items-center space-x-2 text-xs text-slate-5
      <ArrowUpRight size={10} className="text-indigo-600 group-hover:text-cyan-
      <span>{detail}</span>
    </div>
  ))}
</div>

</motion.div>
)}}

```

EXPLANATION DETAILS

Details the header indicators of cards. Represents the card numbers with low-opacity white text (text-white/15 which has 15% opacity), which highlights to 30% (group-hover:text-white/30) on card hover using Tailwind group hover states.



Testimonials.jsx (Draggable Reviews)

src/components/Testimonials.jsx

Responsive, user-draggable sliding track filled with glowing reviews from students.

Lines: 1-44

```
"">use client";

"code-keyword">import { useRef, useState, useEffect } "code-keyword">from "react";
"code-keyword">import { motion } "code-keyword">from "framer-motion";
"code-keyword">import { Star, MessageSquare, Quote, ArrowLeft, ArrowRight } "code-keyword">from "lucide-react";

"code-keyword">const testimonials = [
  {
    id: 1,
    name: "Marcus Vance",
    role: "Senior Creative Developer",
    company: "Vercel",
    avatar: "https://images.unsplash.com/photo-1534528741775-53994a69daeb?q=80&w=200&auto=compress&format=webp",
    quote: "Synapse completely changed how I study. The interactive code sandboxes made coding so much easier.",
    rating: 5,
  },
  {
    id: 2,
    name: "Aria Chen",
    role: "Interaction Designer",
    company: "Framer",
    avatar: "https://images.unsplash.com/photo-1507003211169-0a1dd7228f2d?q=80&w=200&auto=compress&format=webp",
    quote: "The interactive feedback loop is incredible. The AI mentor evaluates layout efficiency in real-time.",
    rating: 5,
  },
  {
    id: 3,
    name: "Devon Reynolds",
    role: "Frontend Engineer",
    company: "Stripe",
    avatar: "https://images.unsplash.com/photo-1500648767791-00dcc994a43e?q=80&w=200&auto=compress&format=webp",
    quote: "I launched my entire creative portfolio while working through the advanced track. The mentorship was invaluable.",
    rating: 5,
  },
  {
    id: 4,
    name: "Elena Rostova",
    role: "Technical Artist",
    company: "Blender",
    avatar: "https://images.unsplash.com/photo-1517841905224-4721686da404?q=80&w=200&auto=compress&format=webp",
    quote: "The 3D environment creation tools are amazing. I've been able to bring my game world to life.",
    rating: 5,
  },
];
```

```
company: "Epic Games",  
avatar: "https://images.unsplash.com/photo-1573496359142-b8d87734a5a2?q=80&w=200&auto=compress",  
quote: "The shader math tracks and layoutId animations are masterfully explained. Truly a masterpiece.",  
rating: 5,  
},  
];
```

EXPLANATION DETAILS

Declares student testimonials array with star ratings, quotes, avatar images, and roles.

Lines: 46-67

```

"code-keyword">export "code-keyword">default "code-keyword">function Testimonials() {
  "code-keyword">const [width, setWidth] = useState(0);
  "code-keyword">const carousel = useRef(null);

  useEffect(() => {
    "code-keyword">if (carousel.current) {
      setWidth(carousel.current.scrollWidth - carousel.current.offsetWidth);
    }
  }, []);

  "code-keyword">const slideLeft = () => {
    "code-keyword">if (carousel.current) {
      carousel.current.scrollBy({ left: -350, behavior: "smooth" });
    }
  };

  "code-keyword">const slideRight = () => {
    "code-keyword">if (carousel.current) {
      carousel.current.scrollBy({ left: 350, behavior: "smooth" });
    }
  };

```

EXPLANATION DETAILS

Initializes variables to track the carousel container dimension bounds using React useRef and useEffect. Calculating the difference between the full scrollable width (scrollWidth) and the visible width (offsetWidth) sets constraints for mobile drag bounds. Defines manual navigator functions (slideLeft, slideRight) using DOM .scrollBy with CSS smooth scrolling animation.

Lines: 108-124

```

"code-comment">{/* Draggable Carousel */}
<motion.div
  ref={carousel}
  className="cursor-grab active:cursor-grabbing overflow-x-auto scrollbar-none flex"
  whileTap={{ cursor: "grabbing" }}
>
  <motion.div
    drag="x"
    dragConstraints={{ right: 0, left: -width }}
    className="flex gap-8"

```

```

>
{testimonials.map((test) => (
  <motion.div
    key={test.id}
    className="w-[300px] sm:w-[380px] shrink-0 select-none rounded-2xl border
  >
    <Quote size={48} className="absolute top-6 right-6 text-slate-900/5 pointer

```

EXPLANATION DETAILS

Implements touch and drag physics. By configuring drag="x" on the child element, users can pull the track horizontally. The dragConstraints restrict the movement, preventing the carousel from being dragged past the boundaries defined by the custom width state.

Lines: 125-156

```

"code-comment">{/* Rating & Quote */}
<div className="space-y-6">
  <div className="flex items-center space-x-1">
    {[...Array(test.rating)].map((_, i) => (
      <Star key={i} size={14} className="fill-indigo-500 text-indigo-500"
    ))}
  </div>
  <p className="text-sm sm:text-base text-slate-700 leading-relaxed font-m
    "{test.quote}"
  </p>
</div>

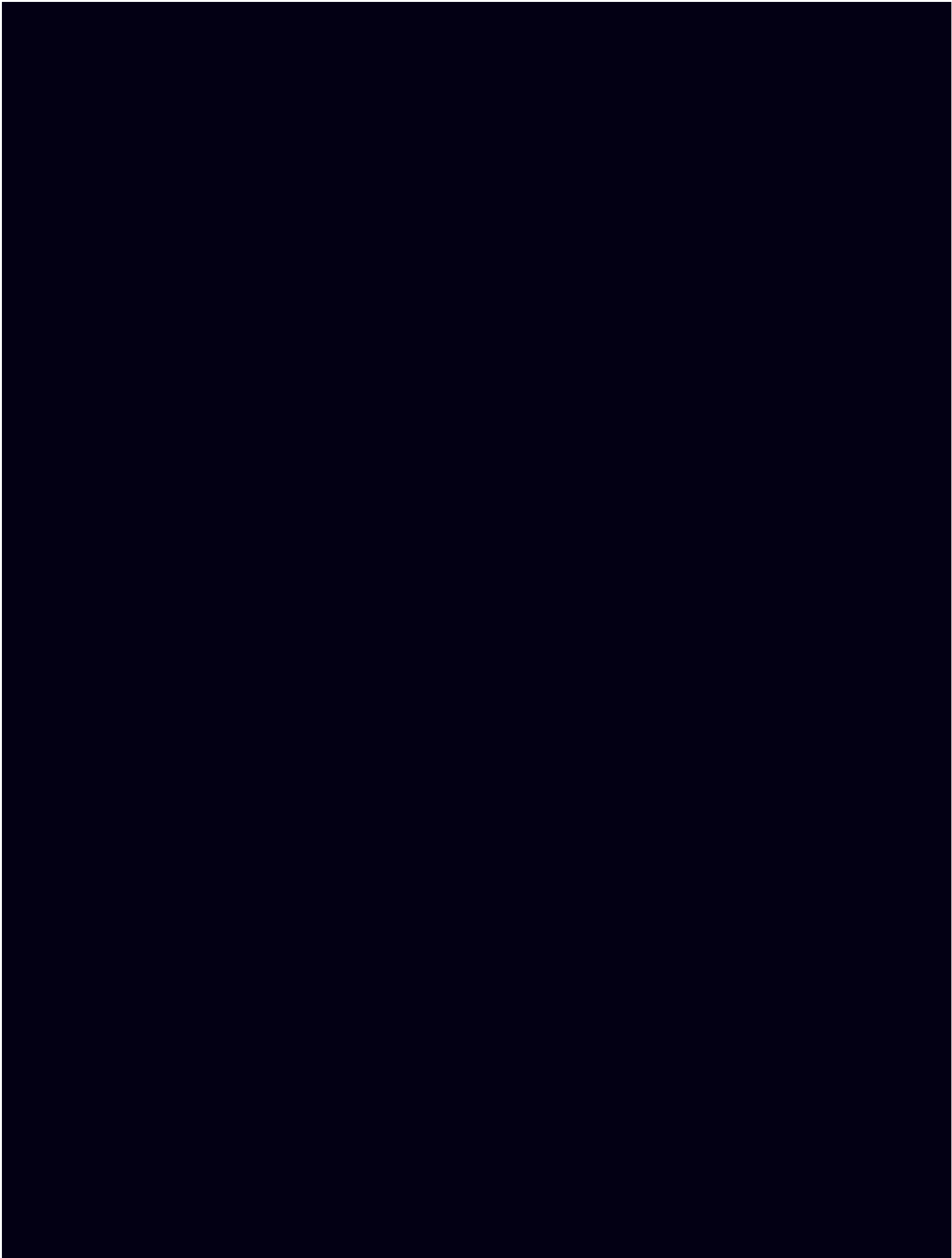
"code-comment">{/* Profile Row */}
<div className="flex items-center space-x-4 mt-8 pt-6 border-t border-slate
  <div className="relative h-12 w-12 overflow-hidden rounded-full border bo
    "code-comment">{/* Fallback to styled initials "code-keyword">if image
    <div className="absolute inset-0 bg-gradient-to-tr ">from-indigo-600 to
      {test.name.charAt(0)}{test.name.split(" ")[1]??.charAt(0)}
    </div>
  </div>
</div>
<div>
  <h4 className="text-sm font-bold font-display text-slate-900">{test.na
  <p className="text-xs text-slate-500">
    {test.role} at <span className="text-slate-700 font-semibold">{test.
  </p>
</div>
</div>

```

```
    </motion.div>  
  )})  
</motion.div>
```

EXPLANATION DETAILS

Loops and renders individual review cards. Each card displays standard gold stars, quotes, student names, and corporate labels.



Pricing.jsx (Subscription Cards)

src/components/Pricing.jsx

Displays the three subscription plans with interactive billing cycle selectors and visual badge tags.

Lines: 1-65

```
"">use client";

"code-keyword">import { useState } "code-keyword">from "react";
"code-keyword">import { motion } "code-keyword">from "framer-motion";
"code-keyword">import { Check, Sparkles } "code-keyword">from "lucide-react";

"code-keyword">const plans = [
  {
    name: "Explorer",
    desc: "Test the waters and explore the basics of code-driven motion design.",
    priceMonthly: 0,
    priceAnnually: 0,
    popular: "code-keyword">false,
    ctaText: "Start Explorer Track",
    gradient: "">from-white to-slate-50/50",
    borderColor: "border-slate-200",
    features: [
      { text: "Access to initial course modules", included: "code-keyword">true },
      { text: "1 Sandbox compiler checkpoint", included: "code-keyword">true },
      { text: "Basic AI code suggestions", included: "code-keyword">true },
      { text: "Interactive community chat", included: "code-keyword">true },
      { text: "Advanced WebGL projects", included: "code-keyword">false },
      { text: "Cryptographic credentials", included: "code-keyword">false },
      { text: "1-on-1 cohort code review", included: "code-keyword">false },
    ],
  },
  {
    name: "Pro Pass",
    desc: "Unlock the complete curriculum, premium sandboxes, and AI mentors.",
    priceMonthly: 29,
    priceAnnually: 19,
    popular: "code-keyword">true,
    ctaText: "Enroll in Pro Access",
    gradient: "">from-indigo-50/20 via-white to-white",
    borderColor: "border-indigo-500",
    features: [
      { text: "Full access to all course tracks", included: "code-keyword">true },
```

```

    { text: "Unlimited browser sandbox renders", included: "code-keyword">true },
    { text: "24/7 AI mentor code audits", included: "code-keyword">true },
    { text: "On-chain cryptographically signed credentials", included: "code-keyword">true },
    { text: "Premium shader & Three.js projects", included: "code-keyword">true },
    { text: "Weekly live cohort review workshops", included: "code-keyword">true },
    { text: "Priority industry recruiter matching", included: "code-keyword">false },
  ],
},
{
  name: "Enterprise",
  desc: "Custom training platforms and scale indicators for agencies & teams.",
  priceMonthly: 99,
  priceAnnually: 79,
  popular: "code-keyword">false,
  ctaText: "Contact Sales",
  gradient: "">from-white to-slate-50/50",
  borderColor: "border-slate-200",
  features: [
    { text: "Everything in Pro Pass included", included: "code-keyword">true },
    { text: "Dedicated company cohort reviews", included: "code-keyword">true },
    { text: "Custom internal training roadmaps", included: "code-keyword">true },
    { text: "Enterprise API key access", included: "code-keyword">true },
    { text: "Dedicated student success manager", included: "code-keyword">true },
    { text: "Recruiter hiring channel alignment", included: "code-keyword">true },
    { text: "Custom SLA agreement support", included: "code-keyword">true },
  ],
},
];

```

EXPLANATION DETAILS

Initializes subscription packages dataset. Pro plan is marked popular and triggers visual borders.

Lines: 90-124

```

<div className="flex items-center justify-center pt-8">
  <div className="relative flex items-center p-1 rounded-full bg-slate-200/50 border">
    <button
      onClick={() => setIsAnnual("code-keyword">false)}
      className={`relative px-4 py-2 text-xs font-bold uppercase transition-colors
        !isAnnual ? "text-slate-900" : "text-slate-500"
      }`}
    >
      {!isAnnual && (

```

```

        <motion.div
          layoutId="pricingToggle"
          className="absolute inset-0 rounded-full bg-white shadow-sm border border-gray-200"
          transition={{ type: "spring", stiffness: 350, damping: 25 }}
        />
      )}
    <span className="relative z-10">Monthly</span>
  </button>
  <button
    onClick={() => setIsAnnual("code-keyword">true)}
    className={`relative px-4 py-2 text-xs font-bold uppercase transition-colors duration-200
      ${isAnnual ? "text-slate-900" : "text-slate-500"}
    `}
  >
    {isAnnual && (
      <motion.div
        layoutId="pricingToggle"
        className="absolute inset-0 rounded-full bg-white shadow-sm border border-gray-200"
        transition={{ type: "spring", stiffness: 350, damping: 25 }}
      />
    )}
    <span className="relative z-10">Yearly Save 30%</span>
  </button>
</div>
</div>
</div>

```

EXPLANATION DETAILS

Renders the cycle toggle controls. Utilizes React state `isAnnual`. Uses a shared `<motion.div layoutId="pricingToggle">` background pill to glide horizontally between cycles, offering a highly polished toggle interaction.

Lines: 127-147

```

<div className="grid grid-cols-1 md:grid-cols-3 gap-8 items-stretch max-w-6xl mx-auto">
  {plans.map((plan) => {
    "code-keyword">const price = isAnnual ? plan.priceAnnually : plan.priceMonthly

    "code-keyword">return (
      <motion.div
        key={plan.name}
        whileHover={{ y: -8 }}
        className={`relative flex flex-col justify-between rounded-2xl border-2 ${

```

```

>
"code-comment">{/* Popular Badge & Glow Border */}
{plan.popular && (
  <>
    <div className="absolute inset-0 rounded-2xl bg-gradient-to-r ">from-indigo-500 to-indigo-600
    <span className="absolute -top-3.5 left-1/2 -translate-x-1/2 inline-flex items-center gap-1">
      <Sparkles size={10} className="animate-pulse" />
      <span>Most Popular</span>
    </span>
  </>
)}

```

EXPLANATION DETAILS

Draws the core pricing cards grid using CSS columns (md:grid-cols-3). Cards adjust their prices in real-time when the cycle switches. Hover actions use Framer Motion whileHover={{ y: -8 }} to lift the card slightly, complementing the standard CSS backdrop-blur panel.

Lines: 149-178

```

<div>
  <h3 className="text-xl font-bold font-display text-slate-900">{plan.name}
  <p className="text-xs text-slate-500 mt-2 min-h-[36px]">{plan.desc}</p>

  "code-comment">{/* Pricing value */}
  <div className="my-8 flex items-baseline">
    <span className="text-5xl font-extrabold font-display text-slate-900">{plan.price}</span>
    <span className="text-5xl font-extrabold font-display text-slate-900 text-right">
      {price}
    </span>
    <span className="text-slate-500 text-sm font-semibold ml-2">/ month</span>
  </div>

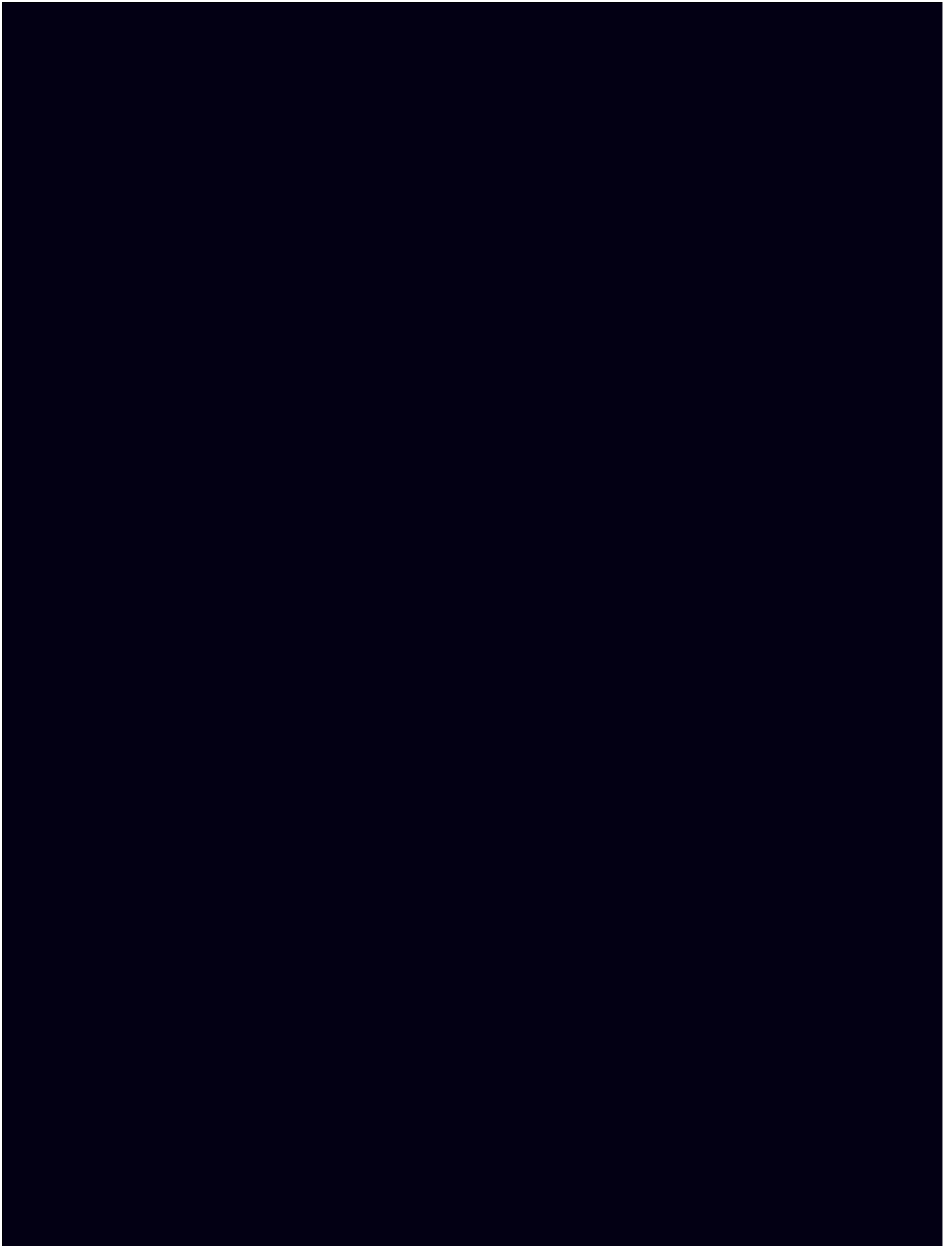
  "code-comment">{/* List of features */}
  <ul className="space-y-4 border-t border-slate-100 pt-8">
    {plan.features.map((feature, fIdx) => (
      <li key={fIdx} className="flex items-start space-x-3 text-sm">
        <div className={`mt-0.5 flex h-4 w-4 shrink-0 items-center justify-center`}>
          feature.included ? "bg-indigo-50 text-indigo-600" : "bg-slate-100 text-slate-500"
        </div>
        <Check size={10} />
      </div>
      <span className={feature.included ? "text-slate-700 font-medium" : "text-slate-500 font-normal"}>
        {feature.text}
      </span>
    )
  )}

```

```
        </span>
      </li>
    )}
  </ul>
</div>
```

EXPLANATION DETAILS

Renders individual plan features. The Check icon matches its color styles to the feature availability (e.g. transparent/slate-700 with cross line-through for unavailable items).



CTA.jsx (Call to Action)

src/components/CTA.jsx

The signup conversion section features an interactive spotlight mouse-tracking background grid.

Lines: 7-27

```
"code-comment">/// Mouse coordinates motion value for glowing background effect
"code-keyword">const mouseX = useMotionValue(0);
"code-keyword">const mouseY = useMotionValue(0);

"code-comment">/// Soft spring effect for custom glow
"code-keyword">const glowX = useSpring(mouseX, { stiffness: 150, damping: 20 });
"code-keyword">const glowY = useSpring(mouseY, { stiffness: 150, damping: 20 });

"code-keyword">function handleMouseMove({ currentTarget, clientX, clientY }) {
  "code-keyword">const { left, top } = currentTarget.getBoundingClientRect();
  mouseX.set(clientX - left);
  mouseY.set(clientY - top);
}

"code-keyword">const backgroundGlow = useMotionTemplate`
  radial-gradient(
    450px circle at ${glowX}px ${glowY}px,
    rgba(99, 102, 241, 0.06),
    transparent 80%
  )
`;
```

EXPLANATION DETAILS

Calculates the current mouse pointer coordinates. useMotionValue registers values that bypass the standard React state render loops for optimal render performance. useSpring smooths the coordinates to eliminate lag. The output is structured inside useMotionTemplate to generate a dynamic CSS radial-gradient spotlight.

Lines: 30-40

```
<section
  onMouseMove={handleMouseMove}
```



```

className="relative py-28 bg-slate-50 overflow-hidden group cursor->default"
>
"code-comment">{/* Dynamic Cursor tracking glow backdrop */}
<motion.div
  className="absolute inset-0 z-0 pointer-events-none opacity-0 group-hover:opacity-100"
  style={{ background: backgroundGlow }}
/>
<div className="absolute inset-x-0 bottom-0 h-px bg-gradient-to-r from-transparent to-transparent"

```

EXPLANATION DETAILS

Applies the radial spotlight glow onto <motion.div> using style attributes. On hover (group-hover:opacity-100), the spotlight follows the mouse pointer within the bounds of the section element.

Lines: 65-88

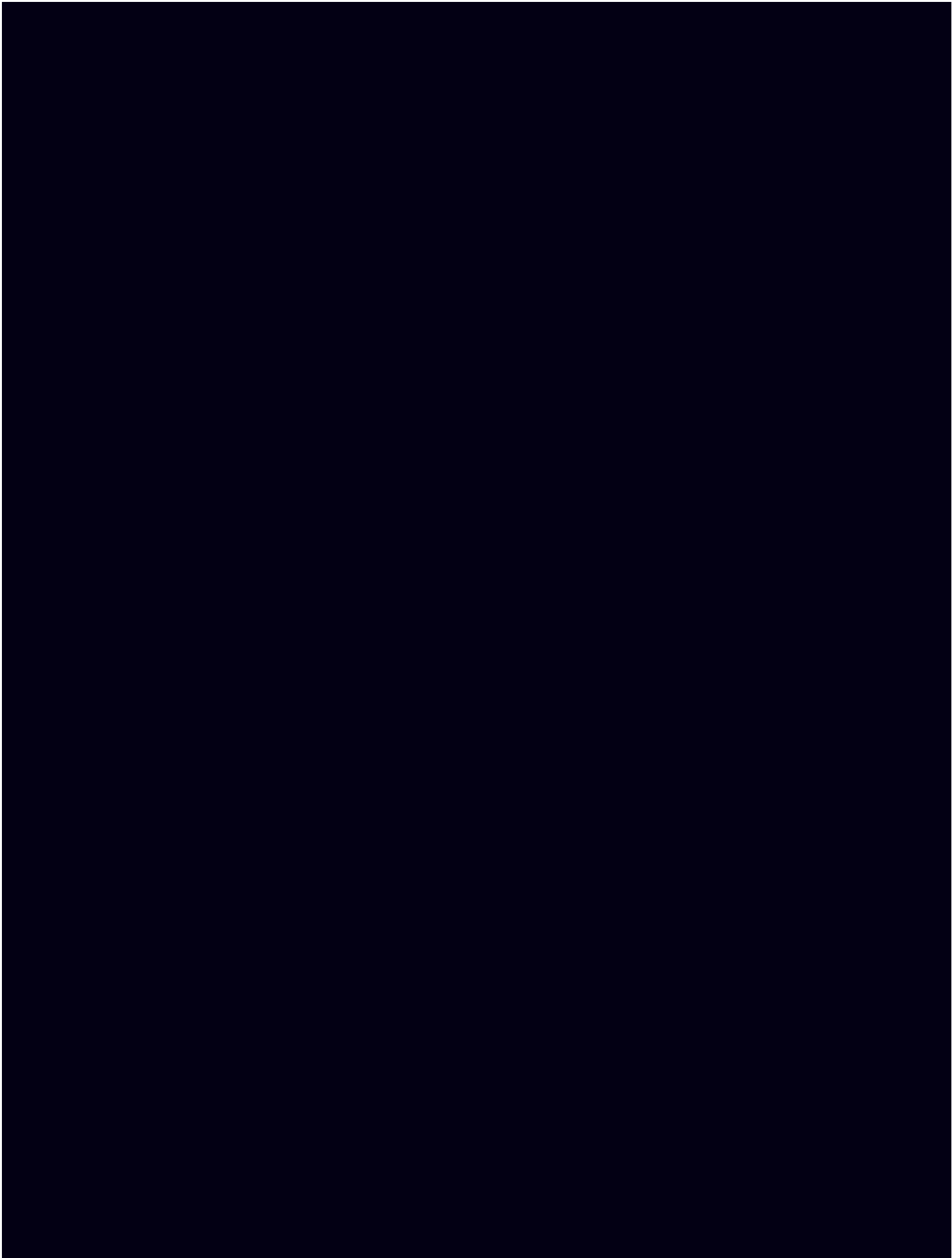
```

"code-comment">{/* Form / CTA button */}
<div className="max-w-md mx-auto pt-4">
  <form onSubmit={(e) => e.preventDefault()} className="flex flex-col sm:flex-row gap-2">
    <input
      type="email"
      required
      placeholder="Enter your email address"
      className="w-full rounded-full border border-slate-200 bg-slate-50 px-6 py-4"
    />
    <motion.button
      type="submit"
      whileHover={{ scale: 1.05 }}
      whileTap={{ scale: 0.95 }}
      className="w-full sm:w-auto shrink-0 inline-flex items-center justify-center"
    >
      <span>Get Started Free</span>
      <ArrowRight size={14} />
    </motion.button>
  </form>
  <p className="text-[10px] text-slate-500 text-left px-4 pt-2">
    No credit card required. Cancel anytime. Pro plan free 7-day trials available.
  </p>
</div>

```

EXPLANATION DETAILS

A flex-based newsletter subscription form container. The input element features border focus animation (focus:border-indigo-500/50) and matches container transitions.



Footer.jsx (Navigation Footer)

src/components/Footer.jsx

Responsive multi-column link list containing inline brand SVGs, credits, and legal anchors.

Lines: 7-24

```
"code-keyword">const TwitterIcon = () => (
  <svg className="w-[18px] h-[18px] fill-current" viewBox="0 0 24 24">
    <path d="M18.244 2.25h3.308l-7.227 8.26 8.502 11.24H16.17l-5.214-6.817L4.99 21.75H1.6817.
  </svg>
);

"code-keyword">const GithubIcon = () => (
  <svg className="w-[18px] h-[18px] fill-current" viewBox="0 0 24 24">
    <path fillRule="evenodd" clipRule="evenodd" d="M12 2C6.477 2 2 6.477 2 12c0 4.42 2.865 8.
  </svg>
);

"code-keyword">const YoutubeIcon = () => (
  <svg className="w-[18px] h-[18px] fill-current" viewBox="0 0 24 24">
    <path d="M23.498 6.163a3.003 3.003 0 00-2.11-2.11C19.518 3.5 12 3.5 12 3.5s-7.518 0-9.388
  </svg>
);
```

EXPLANATION DETAILS

Defines functional components containing inline SVG vectors. By including vectors inline rather than loading static asset routes (e.g. /twitter.svg), standard HTTP requests are minimized, and assets render immediately on load.

Lines: 64-106

```
<footer className="relative bg-white pt-20 pb-12 overflow-hidden border-t border-slate-
  <div className="relative z-10 mx-auto max-w-7xl px-4 md:px-8">

    "code-comment">{/* Upper Link Row */}

    <div className="grid grid-cols-2 md:grid-cols-5 gap-8 pb-16 border-b border-slate-
```

```

"code-comment">{/* Logo column */}
<div className="col-span-2 space-y-4">
  <a href="#" className="flex items-center space-x-2 text-xl font-bold font-disp
    <div className="flex h-8 w-8 items-center justify-center rounded-lg bg-gradie
      <Sparkles size={14} />
    </div>
    <span className="bg-gradient-to-r ">from-slate-900 to-indigo-950 bg-clip-text
      Synapse
    </span>
  </a>
  <p className="text-xs text-slate-600 max-w-sm leading-relaxed">
    Synthesizing engineering precision with high-end creative design. Learn adva
  </p>
</div>

"code-comment">{/* Nav columns */}
{footerLinks.map((col) => (
  <div key={col.title} className="space-y-4">
    <h4 className="text-xs font-bold font-display uppercase tracking-widest text
      {col.title}
    </h4>
    <ul className="space-y-2.5">
      {col.links.map((link) => (
        <li key={link.name}>
          <a
            href={link.href}
            className="text-xs text-slate-600 hover:text-indigo-600 transition-co
          >
            {link.name}
          </a>
        </li>
      ))}
    </ul>
  </div>
))}

</div>

```

EXPLANATION DETAILS

Constructs the link matrix layout. Mobile devices use a 2-column format (grid-cols-2), which automatically shifts to a wider 5-column grid layout (md:grid-cols-5) on desktop screens. The logo column spans 2 units (col-span-2), while columns for learning resources, links, and details consume 1 unit each.

Lines: 108-139

```

"code-comment">{/* Lower Row */}
<div className="flex flex-col sm:flex-row items-center justify-between gap-6 pt-12">

"code-comment">{/* Copyrights */}
<div className="text-[11px] text-slate-500 order-2 sm:order-1 text-center sm:text-1
  <p>© {new Date().getFullYear()} Synapse Academy Inc. All rights reserved.</p>
  <p className="mt-1">
    Developed in flow state using Next.js & Framer Motion.
    <span className="mx-1 text-slate-200">|</span>
    <a href="#" className="hover:text-indigo-600 transition-colors">Privacy Policy<
    <span className="mx-1 text-slate-200">|</span>
    <a href="#" className="hover:text-indigo-600 transition-colors">Terms of Service
  </p>
</div>

"code-comment">{/* Social Links */}
<div className="flex items-center space-x-3 order-1 sm:order-2">
  {socialIcons.map((social, idx) => (
    <motion.a
      key={idx}
      href={social.href}
      whileHover={{ scale: 1.1, y: -2 }}
      whileTap={{ scale: 0.95 }}
      className={`flex h-10 w-10 items-center justify-center rounded-xl border border
    >
      {social.icon}
    </motion.a>
  )))}
</div>

</div>

```

EXPLANATION DETAILS

The footer credit bar. Renders dynamic system years using `new Date().getFullYear()`. The social media anchors use hover motions (`whileHover={{ scale: 1.1, y: -2 }}`) to lift slightly on hover.

3. Tailwind Utility Cheat Sheet

Fast reference dictionary of styling classes applied in the workspace.

CATEGORY	TAILWIND CLASS PATTERNS	CSS EQUIVALENTS / PURPOSE
Flexbox & Grid Layouts	<code>flex, grid, items-center, justify-between, grid-cols-1 md:grid-cols-3</code>	Sets containers to flex or grid layouts. Directs positioning along parent axes (e.g. centering cross-axis items or spacing child items evenly). Grid patterns define grid column divisions responsive to screen resolutions.
Sizing & Layout Wrappers	<code>min-h-screen, w-full, h-[500px], mx-auto, max-w-7xl</code>	Defines dimension parameters. min-h-screen sets minimum height to 100vh. Brackets represent absolute arbitrary overrides. mx-auto max-w-7xl wraps content inside page centers with side-margin buffers.
Interactive Effects	<code>transition-all, duration-300, hover:scale-105, group-hover:opacity-100</code>	Sets up CSS transition filters, speeds, and triggers state transforms when hovering over targets or group elements.
Glassmorphism & Shadows	<code>backdrop-blur-md, bg-slate-950/45, border-white/5, shadow-2xl</code>	Applies glass aesthetic styles. Combines transparent overlay colors with CSS background blurs and soft shadow layers.